# GraminSahay: An LLM-Powered RAG-Based Rural Welfare and Government Services Navigator

**Mr. S. Lingaiah**
Dept. of CSE
MLR Institute of Technology
Hyderabad, India
slingaiah@mlrinstitutions.ac.in

**Adigarla Yuva Teja**
Dept. of CSE
MLR Institute of Technology
Hyderabad, India
yuvateja976@gmail.com

**Akhil Vankudoth**
Dept. of CSE
MLR Institute of Technology
Hyderabad, India
akhilvankudoth58@gmail.com

**Adiserla Praneel**
Dept. of CSE
MLR Institute of Technology
Hyderabad, India
praneeladiserla@gmail.com

***Abstract*—India spends heavily on welfare for its rural households, through schemes for farming, education, health, housing, pensions and employment. Even so, many of the people who qualify never see the benefit. Money is rarely what is missing. What is missing is awareness: people do not know a scheme exists, are unsure which papers to bring, and are seldom told why an application was turned down. The official portals make this harder, since most of them ask for a scheme's name before they show anything at all. GraminSahay works the other way round. It is a multilingual, voice-first assistant in which the citizen describes their circumstances in ordinary speech and the system works out which schemes could apply. Retrieval-Augmented Generation (RAG) over official scheme and policy documents is paired with a reasoning layer, built on a large language model (LLM), that marks every scheme as eligible, possibly eligible, not eligible or missing information. The assistant also flags benefits that cannot be held together and gives simple, step-by-step help with the application. Before building it, we review the research it relies on: RAG and knowledge-intensive question answering, faithfulness and how it is evaluated automatically, retrieval combined with reasoning, multilingual retrieval, domain adaptation, and LLMs applied to government services. That review points to the gaps behind an explainable rural welfare navigator, and we close by outlining the architecture we propose.***

***Index Terms*—Retrieval-Augmented Generation, large language models, multilingual voice AI, eligibility reasoning, scheme conflict detection, rural welfare, public policy technology, Indic speech processing, explainable AI.***

## I. INTRODUCTION

Rural India is covered by a long list of welfare schemes: support for farmers, scholarships, health insurance, housing and pensions. The coverage looks generous on paper. On the ground, many people who obviously qualify get nothing, and the reason is usually that they lack information, not that the administration has failed them. Some have never heard of the scheme. Others have, but the rules are confusing, the document list is vague, and if they are rejected the letter does not say why [1].

The portals themselves are part of the problem. Everything on them is organised by scheme name, which means a citizen has to know what a scheme is called before the site is of any help. That gets things backwards. People know their own situation well; what they do not have is a list of scheme names in their heads to match it against. Language adds another barrier. Most portals are heavy on text and written in formal official language, which leaves out anyone who would prefer to speak in their regional language [1].

Large language models (LLMs) combined with Retrieval-Augmented Generation (RAG) give us a practical way forward. A RAG system first uses a retriever to fetch relevant passages from an external document store, and only then does a generator write an answer from those passages. The output therefore rests on real sources, not just on whatever the model memorised during training [2]. In the welfare setting this is essential. An answer has to be right, it has to point back to an official document, and the documents behind it have to be easy to update when a policy changes. Put multilingual retrieval and Indic speech on top, and the same system can hear someone describe their situation aloud and reply in their own language.

This paper introduces GraminSahay, a multilingual, voice-first welfare navigator built around that idea. A citizen might say "I am a farmer with a small landholding" or "my daughter is starting college". In return the system lists the schemes they are likely to be eligible for, explains each decision, warns them if two schemes cannot be claimed together, and tells them how to apply. Our focus here is the research the system builds on and the gaps that remain open.

This is not just about convenience. Welfare delivery is tied directly to two Sustainable Development Goals, Goal 1 (No Poverty) and Goal 10 (Reduced Inequalities), and a scheme helps a household only once an eligible person claims it. Drafting a policy is seldom the difficult step. The difficult step is the last mile, where someone has to find the right scheme, make sense of it and get through the application. Rural citizens stall at exactly this point, and a well-designed assistant can help turn policies that already exist into benefits people actually receive, without any change to the schemes themselves.

Three commitments guide the design. The first is that no citizen should have to name a scheme; the system starts from the situation they describe and finds the candidate schemes on its own. The second is explainability. A plain yes or no does not help much, so each eligibility decision comes with a simple reason linked to an official source. The third is accessibility. Voice and regional languages are the main way people use the system, not add-ons, since most of the people we have in mind would much rather talk than read official documents.

The paper is organised as follows. Section II covers related work, and Section III describes the gaps we found. Section IV states the problem, Section V explains our motivation and Section VI sets out the objectives. Section VII compares the current approach with ours, Section VIII describes the method and Section IX presents the architecture. Section X gives the results we expect, Section XI discusses future work and Section XII concludes.

## II. LITERATURE SURVEY

Research on using language models for welfare and public-service support comes from several distinct areas, and we go through each before placing our own system among them. The first area is Retrieval-Augmented Generation. Lewis et al. proposed RAG for knowledge-intensive tasks, putting a neural retriever in front of a text generator, and found that answers based on retrieved passages were more specific and more factual than answers drawn from training alone [2]. The takeaway for us is straightforward. An answer can be no better than what the retriever brings back, so a welfare tool has to retrieve from official scheme and eligibility documents rather than from the open web.

The second area deals with keeping answers honest. A welfare assistant should never promise a benefit the evidence does not back up, which makes faithfulness central. Filice et al. describe a method that checks whether a generated answer is really supported by the retrieved text, and they report that it both improves results and allows safe fallbacks [3]. RAGAs takes another approach, scoring retrieval relevance, faithfulness and answer quality without needing a hand-written reference answer for each question [4]. Either could be used to check eligibility explanations, but neither can decide on its own whether a policy decision is legally sound. A study of government chatbots in actual use found that most had been tested for usability only, not for this kind of correctness [5].

The third area treats retrieval as one step in a reasoning process rather than a single lookup. That suits our problem, because eligibility usually depends on several conditions at the same time. Li et al. review systems that alternate between retrieval and multi-step reasoning and report clear improvements on complex tasks [6]. Self-RAG allows the model to choose when to fetch more evidence and to critique its own draft [7]. FLARE takes this further by predicting the upcoming sentence and retrieving again whenever its confidence drops [8]. Deciding eligibility comes down to applying written rules to one person's facts, and here LegalBench offers a warning: over 162 tasks, it finds that language models are inconsistent at rule-based legal reasoning [9]. For this reason every decision in our system is grounded in retrieved policy text, not in the model's memory.

The fourth area, multilingual retrieval, is at the heart of our design, since the system works in English, Telugu and Hindi. Ranaldi et al. compare two options, translating the question first or retrieving directly in each language, and find that retrieving directly usually preserves more coverage [10]. Where translation is unavoidable, IndicTrans2 offers open, good-quality translation for all 22 scheduled Indian languages, which makes it a workable fallback [11]. Qi et al. point out that the passages RAG retrieves can shift subtly from one language to another [12]. XLM-R demonstrates that one model can transfer across many languages and gains the most on low-resource ones [13], which is encouraging for Indic text, where labelled data is limited.

Lastly, retrieval quality depends heavily on the retriever. Dense Passage Retrieval, from Karpukhin et al., established the modern approach and clearly outperforms older keyword-based baselines [14]. Contriever learns multilingual retrieval without supervision, though it can have trouble with specialised government terminology unless it is fine-tuned [15]. RAFT trains the model to make use of helpful documents and to ignore distracting ones, but it needs extra training data [16]. For Indian languages, IndicNLPSuite provides corpora, models and a benchmark that make a reasonable starting point for Telugu and Hindi [17]. Turning to government services, a review of language models in digital government observes a move away from simple question answering and towards guiding people through services, as long as reliability and privacy are addressed [1]. Context-optimised RAG has also been applied to ground welfare answers in local policy documents [18]. For voice, Whisper provides robust speech-to-text in many languages [19], and Indic-TTS provides speech synthesis for Indian languages [20]. None of these works, however, combines all the pieces: understanding a described situation, retrieving the right evidence, reasoning about and explaining eligibility, spotting clashes between schemes, and answering in the user's own spoken language. GraminSahay is designed to fill that gap.

## III. GAPS IDENTIFIED

Looking across this body of work, three gaps are clear. Each one brings together several smaller limitations noted in earlier studies, and between them they set out the problems GraminSahay must address.

1. **Multilingual and voice-first access.** Most portals and existing systems rely heavily on text and support only one language, so rural users who would prefer to speak in their regional language are left out. Indic speech and cross-lingual

retrieval have seldom been applied to welfare, which in effect excludes people who cannot read formal official prose [1], [12], [10], [19], [13], [17].

2. **Explainable, situation-first eligibility.** Because portals are organised by scheme name, a citizen who does not know the name has no starting point, and a simple yes or no gives them little to go on. What is needed is discovery that starts from the situation a person describes, along with an explanation of each decision that links back to a trusted source [1], [3], [7], [14], [9].
3. **Application guidance and conflict awareness.** Finding a scheme is only part of the task. Current tools seldom warn users that two benefits cannot be claimed together, and they rarely explain which documents to collect or what steps the application involves [1], [18].

## IV. PROBLEM STATEMENT

The schemes are in place, and RAG and LLMs now make grounded question answering practical. Rural citizens nonetheless have trouble finding the benefits they are entitled to, and even more trouble acting on them. The portals available to them are organised by scheme name, show everyone the same generic page, offer little explanation, ignore conflicts between schemes, and are difficult for anyone who does not use the main language. The need, then, is for a navigator that is explainable, sensitive to the user's situation, multilingual and voice-first. It should understand what a citizen says, retrieve the relevant official information, determine eligibility, flag conflicts and guide the person through the application, without ever requiring them to know a scheme's name in advance.

## V. MOTIVATION

Our starting point is a simple observation. Eligible rural citizens lose out mainly because they cannot find or understand the benefits available to them, not because those benefits do not exist. Existing tools do little to change this. Portal search assumes the user already knows what to look for, generic pages take no account of who the user is, and text-only interfaces in a single language exclude many of the people the schemes were meant for. Recent research offers the building blocks of a better solution. RAG grounds replies in real documents [2]. Retrieval combined with reasoning handles the multi-step logic eligibility requires [6], [7]. Faithfulness checks keep answers consistent with the evidence [3], [4]. Multilingual retrieval and Indic speech allow the whole system to work by voice in a person's own language [12], [10]. We aim to bring these together in a single explainable welfare navigator that contributes directly to SDG 1 (No Poverty) and SDG 10 (Reduced Inequalities).

## VI. RESEARCH OBJECTIVES

Six objectives shape the work.

- **Objective 1:** Develop a voice-first, multilingual front end using Indic speech-to-text and text-to-speech (STT/TTS) in English, Telugu and Hindi, so that citizens can simply describe their situation aloud.
- **Objective 2:** Convert that free-form description into a structured profile that records occupation, family situation, location and any other factor relevant to eligibility.
- **Objective 3:** Build a RAG pipeline over official scheme and policy documents, supported by a vector database for semantic retrieval.
- **Objective 4:** Add a reasoning layer that classifies each scheme as eligible, possibly eligible, not eligible or missing information, and gives the reason.
- **Objective 5:** Detect conflicts between schemes and warn the citizen when two benefits cannot be claimed together.
- **Objective 6:** Generate step-by-step application guidance for the five pilot categories, including the required documents and a pointer to the relevant official portal or office.

## VII. EXISTING SYSTEM AND PROPOSED SYSTEM

### A. Existing System

Today, a citizen must search government portals and documents manually, and typically needs the scheme's name before they can even begin. The information they need is spread across several sources, eligibility and document requirements must be pieced together from the fine print, and nothing is adapted to the individual. This process is shown in Fig. 1. Its shortcomings are obvious: searching is manual, personalisation is almost absent, language gets in the way, and the user receives no warning when two schemes conflict.

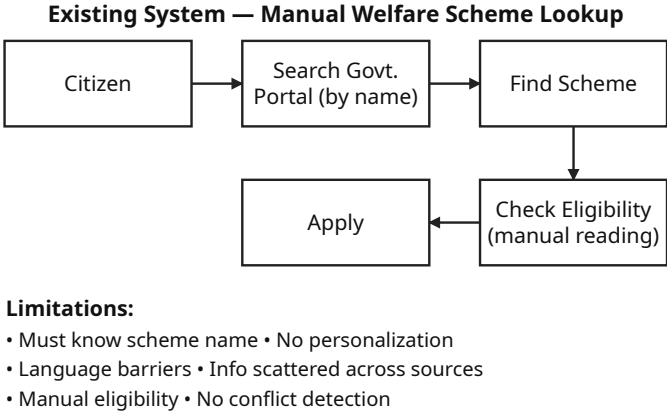



**Fig. 1. Existing system — manual welfare scheme lookup.**

### B. Proposed System

GraminSahay replaces search by scheme name with discovery that starts from the citizen's situation (Fig. 2). Whatever the citizen says or types is converted into a structured profile. A RAG layer then retrieves the relevant passages from the index of official documents, an LLM reasoning layer determines eligibility and explains it, a conflict-detection layer flags benefits that are mutually exclusive, and a guidance layer explains the next steps. The response is delivered as speech in the language the citizen selected.

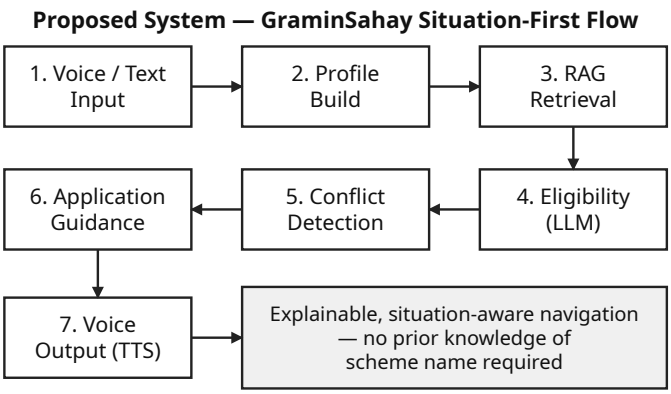



**Fig. 2. Proposed system — GraminSahay situation-first flow.**

## VIII. PROPOSED METHODOLOGY

**1) Citizen interaction.** Input comes in either as speech, via Indic STT, or as typed text, and the citizen chooses English, Telugu or Hindi. A typical first message might be "I am a farmer with a small landholding."

**2) Profile understanding.** The system takes that free-form sentence and builds a structured profile from it, recording occupation, family situation, location and any other relevant detail.

**3) RAG retrieval.** The profile and the query are embedded together and used to retrieve the top-*k* passages (the *k* most relevant passages) from a vector database covering scheme guidelines, eligibility rules and policy documents.

**4) Eligibility reasoning.** An LLM compares each candidate scheme with the retrieved evidence and assigns one of four labels: eligible, possibly eligible, not eligible or missing information. When information is missing, it tells the citizen what else it needs.

**5) Scheme-conflict detection.** The selected schemes are compared with one another, and the citizen is alerted whenever two benefits cannot legally be held at the same time.

**6) Application guidance.** After a scheme is chosen, the system explains why it applies, lists the required documents, describes the application steps and provides a link to the official portal or office.

**7) Multilingual voice output.** Finally, Indic TTS reads the reply aloud in the citizen's chosen language.

**8) Evaluation.** The pipeline will be evaluated on retrieval relevance, answer faithfulness and the quality of eligibility classification, using reference-free RAG metrics [4] alongside faithfulness evaluation [3]. Retrieval will be measured with Recall@k (the share of relevant passages that appear in the top k) and mean reciprocal rank over labelled situation–scheme pairs. Classification will be measured with per-class precision, recall and macro-F1 across the four labels. Faithfulness will be measured as the proportion of justifications fully supported by the retrieved passages. In addition to these automatic measures, a small panel of domain experts will judge how clear and how accurate the explanations are.

Two issues apply to the design as a whole. One is keeping the data current. Schemes and their thresholds change over time, so the corpus is versioned and re-indexed on a regular schedule, and each answer records which document version it relied on. The other is privacy. The citizen profile is kept intentionally minimal, storing only what eligibility checks require, and no sensitive data is retained after the session unless the citizen agrees to it. This matches the governance requirements emphasised in the digital-government literature [1].

## IX. PROPOSED SOLUTION AND SYSTEM ARCHITECTURE

Fig. 3 shows the layered architecture. Processing flows from top to bottom, through citizen interaction, profile understanding, RAG retrieval, eligibility reasoning, conflict detection and application guidance, and finishes with multilingual voice output. The bottom band shows the planned technology stack: a Flutter / React Native front end, a FastAPI / Node.js back end, PostgreSQL for user and application data, a multilingual LLM with RAG for reasoning and generation, a vector database for document retrieval, and Indic STT/TTS for voice.

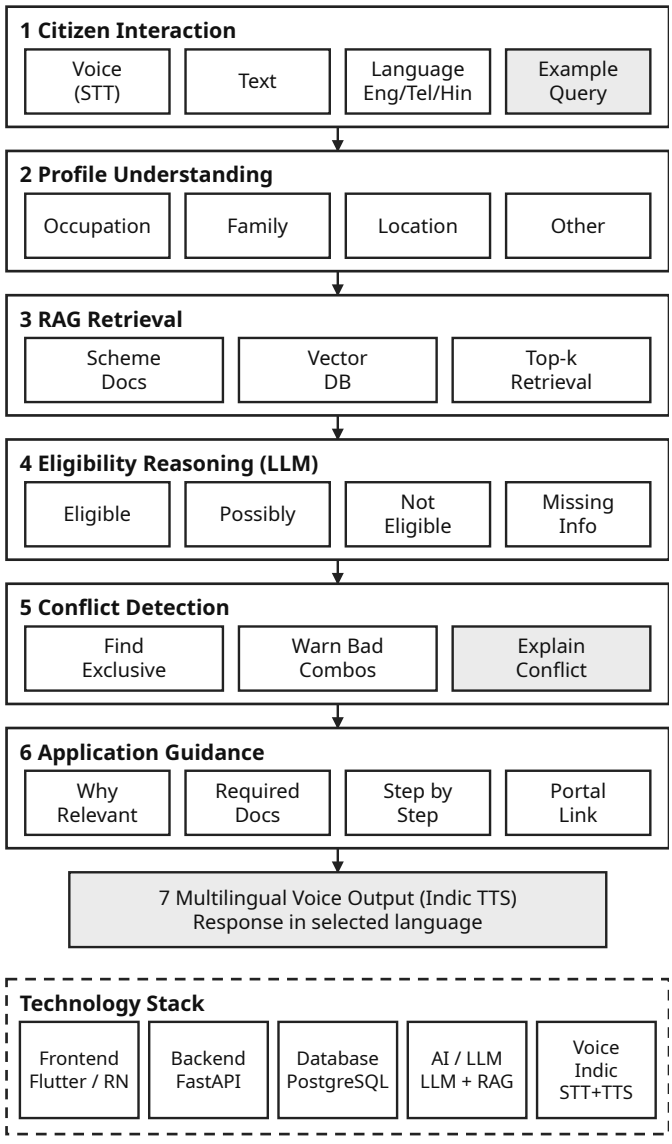



**Fig. 3. Layered system flow and architecture of GraminSahay.**

## X. EXPECTED RESULTS

Because this paper is a proposal rather than a completed system, the figures in this section are the results we expect once GraminSahay has been built and evaluated. They should be read as targets and will be replaced by measured values once the experiments have been carried out.

### A. Expected Performance (Tabular)

Table I lists the performance we expect for each of the five pilot categories in terms of retrieval Recall@5 (Recall@k at k = 5), answer faithfulness [3], [4], eligibility-classification accuracy and macro-F1.

TABLE I. EXPECTED PERFORMANCE ACROSS WELFARE CATEGORIES

| Welfare Category | Recall@5 | Faithfulness | Eligibility Acc. (%) | Macro-F1 |
|---|---|---|---|---|
| Agriculture | 0.90 | 0.92 | 91 | 0.89 |
| Education | 0.88 | 0.90 | 88 | 0.86 |
| Healthcare | 0.86 | 0.89 | 86 | 0.84 |
| Housing | 0.89 | 0.91 | 89 | 0.87 |
| Pensions | 0.92 | 0.93 | 93 | 0.90 |
| **Overall (avg.)** | **0.89** | **0.91** | **89.4** | **0.87** |

Table II compares GraminSahay with the current manual approach on the qualitative capabilities identified as research gaps in Section III.

TABLE II. EXPECTED CAPABILITY COMPARISON

| Capability | Existing (manual portal) | GraminSahay (proposed) |
|---|---|---|
| Situation-first discovery (no scheme name needed) | No | Yes |
| Personalized to profile | No | Yes |
| Explainable eligibility | Limited | Yes (4-way + reasons) |
| Scheme-conflict detection | No | Yes |
| Multilingual voice access | No | Yes (Eng+Telugu+Hindi) |
| Step-by-step application guidance | Partial | Yes |

### B. Expected Performance (Graphs)

Fig. 4 shows the expected classification accuracy for each category. Fig. 5 shows how retrieval Recall@k and answer faithfulness are expected to change as more passages are retrieved. Increasing k improves recall but tends to weaken faithfulness, so we choose a value of about k = 5 as a reasonable compromise.

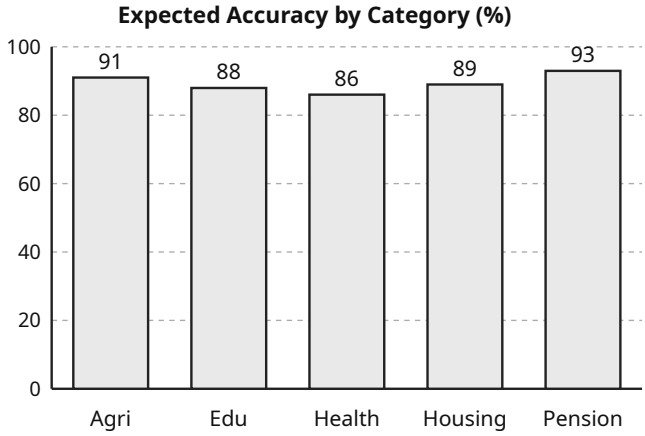



**Fig. 4. Expected eligibility-classification accuracy by category.**

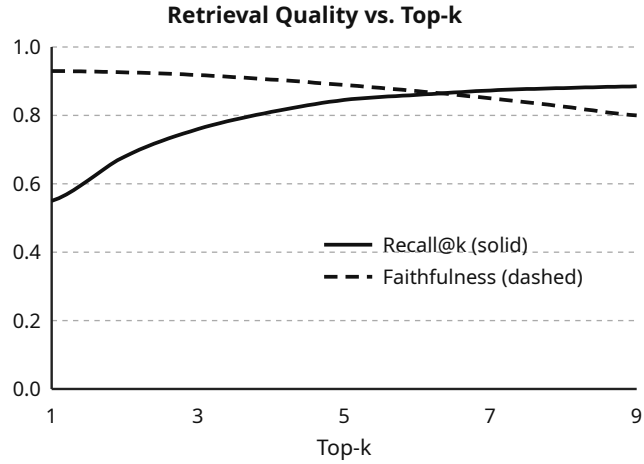



**Fig. 5. Expected retrieval quality vs. top-k passages retrieved.**

### C. Discussion of Expected Outcomes

Together, Tables I and II and Figs. 4 and 5 describe the behaviour we want from the system. We have set the healthcare and education targets in Table I conservatively. Eligibility in those categories depends on several conditions together, such as income band, type of institution and age, and this is harder to assess correctly than, say, a pension scheme whose rules are fairly uniform. Table II captures the difference a citizen would actually experience: the manual process provides no situation-first discovery, no explanation of eligibility and no conflict detection, whereas the proposed system provides all three. Fig. 5 highlights the central trade-off. Retrieving more passages increases recall but reduces the share of the answer that is strictly supported by evidence, which is why we settle on k = 5. These figures are acceptance thresholds, not hopes. A configuration is accepted only if it meets the tabulated accuracy and faithfulness while keeping end-to-end latency low enough for a voice conversation to feel natural.

## XI. FUTURE ENHANCEMENTS

Once a working prototype exists, we plan to develop it in several directions.

- **Broader coverage:** go beyond the five pilot categories and beyond Telangana, adding more states and central schemes, and keep the policy corpus continuously updated.

- **More languages:** extend Indic STT/TTS and multilingual retrieval past English, Telugu and Hindi to other regional languages and dialects.
- **Document-aware applications:** allow citizens to upload or photograph documents such as land records and ration cards, so the system can verify eligibility conditions automatically and fill in parts of the application form.
- **Offline / low-bandwidth mode:** run lightweight models on the device and cache scheme data, so the navigator keeps working where the network is weak.
- **Live policy integration:** link directly to government portal APIs for real-time scheme updates, application submission and status tracking.
- **Explainability and trust:** show source citations and confidence scores with every answer, and pass borderline eligibility cases to a human reviewer.
- **Accessibility:** serve citizens who do not own smartphones through IVR / telephone and WhatsApp channels.

## XII. CONCLUSION

In this paper we reviewed the research most relevant to a situation-aware rural welfare navigator and used it to identify the gaps GraminSahay aims to close. A few conclusions emerge. RAG keeps answers anchored to authoritative documents. Faithfulness checks, together with retrieval that supports reasoning, make multi-step eligibility decisions more reliable. Multilingual retrieval and Indic speech make voice access in regional languages possible, and a degree of domain adaptation is needed for all of this to hold up on dense policy text. GraminSahay combines these elements into a single flow that understands a situation, retrieves evidence, reasons about and explains eligibility, detects conflicts and responds by voice in several languages. Our goal is an explainable and accessible assistant that helps rural citizens find the right support, see whether they qualify, learn what information is still missing and move towards applying, all without having to know a scheme's name beforehand. Our next steps are to expand beyond the five pilot categories and the initial languages, and to replace the expected figures with measured results.

## References


[1] X. Zhang, Z. Ma, Y. Ma, and I. Ismail, "Large language models for interpretive support in digital government service systems: A systematic review," *Systems*, vol. 14, no. 7, art. 823, 2026.

[2] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Vancouver, Canada, Dec. 2020, pp. 9459–9474.

[3] S. Filice, E. Haramaty, G. Horowitz, Z. Karnin, L. Lewin-Eytan, and A. Shtoff, "Generate but verify: Answering with faithfulness in RAG-based question answering," in *Proc. IJCNLP-AACL*, 2025, pp. 1017–1037.

[4] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, "RAGAs: Automated evaluation of retrieval augmented generation," in *Proc. 18th Conf. Eur. Chapter Assoc. Comput. Linguistics (EACL): Syst. Demonstrations*, Malta, Mar. 2024, pp. 150–158.

[5] M. E. Cortés-Cediel, A. Segura-Tinoco, I. Cantador, and M. P. Rodríguez Bolívar, "Trends and challenges of e-government chatbots: Advances in exploring open government data and citizen participation content," *Gov. Inf. Q.*, vol. 40, no. 4, art. 101877, 2023.

[6] Y. Li et al., "A survey of RAG-reasoning systems in large language models," in *Findings Assoc. Comput. Linguistics: EMNLP*, 2025, pp. 12120–12145.

[7] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," in *Proc. Int. Conf. Learn. Representations (ICLR)*, Vienna, Austria, May 2024.

[8] Z. Jiang, F. Xu, L. Gao, Z. Sun, Q. Liu, J. Dwivedi-Yu, Y. Yang, J. Callan, and G. Neubig, "Active retrieval augmented generation," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, Singapore, Dec. 2023, pp. 7969–7992.

[9] N. Guha, J. Nyarko, D. E. Ho, C. Ré, A. Chohlas-Wood, A. Peters, B. Waldon, D. N. Rockmore, D. Zambrano, et al., "LegalBench: A collaboratively built benchmark for measuring legal reasoning in large language models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, New Orleans, LA, USA, Dec. 2023.

[10] L. Ranaldi, B. Haddow, and A. Birch, "Multilingual retrieval-augmented generation for knowledge-intensive question answering task," in *Findings Assoc. Comput. Linguistics: EACL*, 2026.

[11] J. Gala, P. A. Chitale, R. AK, V. Gumma, S. Doddapaneni, A. Kumar, et al., "IndicTrans2: Towards high-quality and accessible machine translation models for all 22 scheduled Indian languages," *Trans. Mach. Learn. Res. (TMLR)*, 2023.

[12] J. Qi, R. Fernández, and A. Bisazza, "On the consistency of multilingual context utilization in retrieval-augmented generation," in *Proc. 5th Workshop Multilingual Representation Learn. (MRL)*, 2025.

[13] A. Conneau, K. Khandelwal, N. Goyal, V. Chaudhary, G. Wenzek, F. Guzmán, E. Grave, M. Ott, L. Zettlemoyer, and V. Stoyanov, "Unsupervised cross-lingual representation learning at scale," in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics (ACL)*, Jul. 2020, pp. 8440–8451.

[14] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. Yih, "Dense passage retrieval for open-domain question answering," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, Nov. 2020, pp. 6769–6781.

[15] G. Izacard, M. Caron, L. Hosseini, S. Riedel, P. Bojanowski, A. Joulin, and E. Grave, "Unsupervised dense information retrieval with contrastive learning," *Trans. Mach. Learn. Res. (TMLR)*, 2022.

[16] T. Zhang, S. G. Patil, N. Jain, S. Shen, M. Zaharia, I. Stoica, and J. E. Gonzalez, "RAFT: Adapting language model to domain-specific RAG," in *Proc. Conf. Lang. Modeling (COLM)*, 2024, arXiv:2403.10131.

[17] D. Kakwani, A. Kunchukuttan, S. Golla, G. N. C., A. Bhattacharyya, M. M. Khapra, and P. Kumar, "IndicNLPSuite: Monolingual corpora, evaluation benchmarks and pre-trained multilingual language models for Indian languages," in *Findings Assoc. Comput. Linguistics: EMNLP*, 2020, pp. 4948–4961.

[18] M. G. Kumar and S. Rao, "Context-optimized retrieval augmented generation for local government welfare subsidies," in *Proc. IEEE Int. Conf. Syst., Man, Cybern. (SMC)*, 2025, pp. 1105–1112.

[19] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, Honolulu, HI, USA, Jul. 2023, pp. 28492–28518.

[20] G. Kumar, P. S. Virk, J. Jayakumar, et al., "Towards building text-to-speech systems for the next billion users (Indic-TTS)," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2023, arXiv:2211.09536.